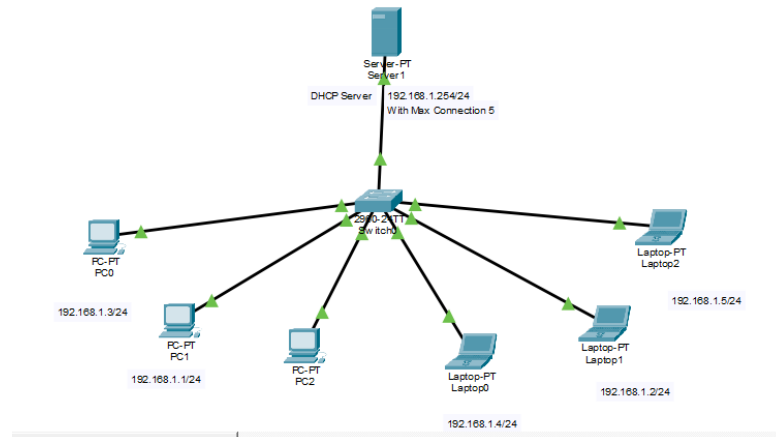


LAB-3

1. To perform a DHCP lab using Cisco Packet Tracer

Steps to Perform this Lab:



Step 1: Add Devices

Open Cisco Packet Tracer.

Add:

- 1 × Server
- 1 × Switch (2960)
- 3 × PCs
- 3 × Laptops

Connect all devices to the switch using Copper Straight-Through cables.

Step 2: Configure the Server IP

Server → Config → FastEthernet0 → Static

Field	Value
IP Address	192.168.1.254
Subnet Mask	255.255.255.0

Server → Config → Setting →

Default Gateway	192.168.1.254 (or leave blank if only one LAN)
DNS	8.8.8.8 / 192.168.1.254

Close the window.

Step 3: Configure DHCP

Server → Services → DHCP

Turn **Service = On**

Fill in:

Field	Value
Pool Name	serverPool
Default Gateway	192.168.1.254
DNS Server	192.168.1.254
Start IP Address	192.168.1.1

Field	Value
Subnet Mask	255.255.255.0
Maximum Number of Users	5
Click Save.	

Step 4: Configure DNS
Server → Services → DNS
Turn DNS Service = On.
Enter:

Field	Value
Name	anudeep.com
Type	A Record
Address	192.168.1.254
Click Add.	

Step 5: Configure HTTP
Server → Services → HTTP
Set HTTP = On.
<html>
<h1>Element heading</h1>
<h2>Heading 2</h2>
</html>

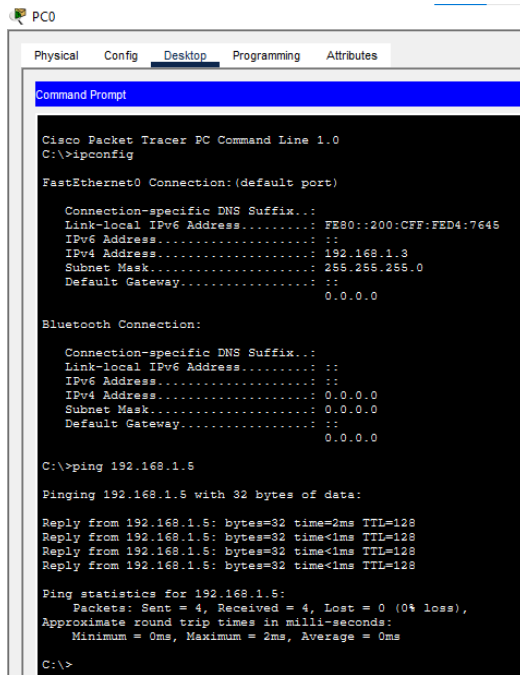
Click **Save**.

Step 6: Configure All Clients
For **each PC/Laptop**:
Desktop → IP Configuration
Click **DHCP**.
The devices will automatically receive:

Device	IP
PC0	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
Laptop0	192.168.1.4
Laptop1	192.168.1.5

Laptop2 will not receive an IP because the DHCP pool allows only **5 users**.

From PC 0 ➔ ping 192.168.1.5



```
PC0
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::200:CFF:FED4:7645
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.3
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .:
                                0.0.0.0

C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

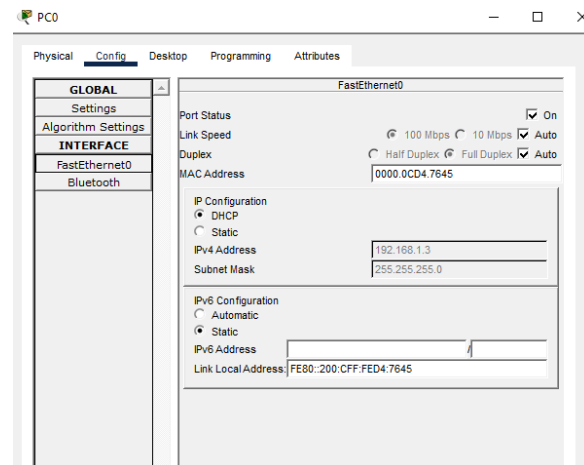
Reply from 192.168.1.5: bytes=32 time=2ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

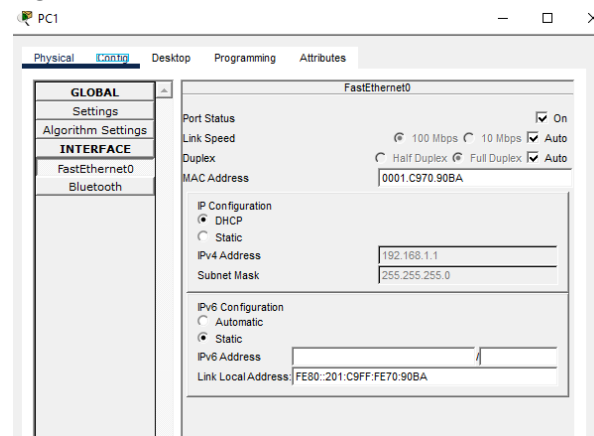
C:\>
```

Successfully Connected.

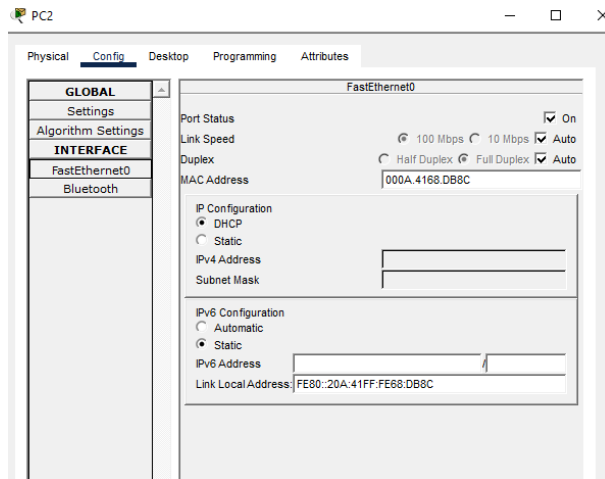
PC0



PC1

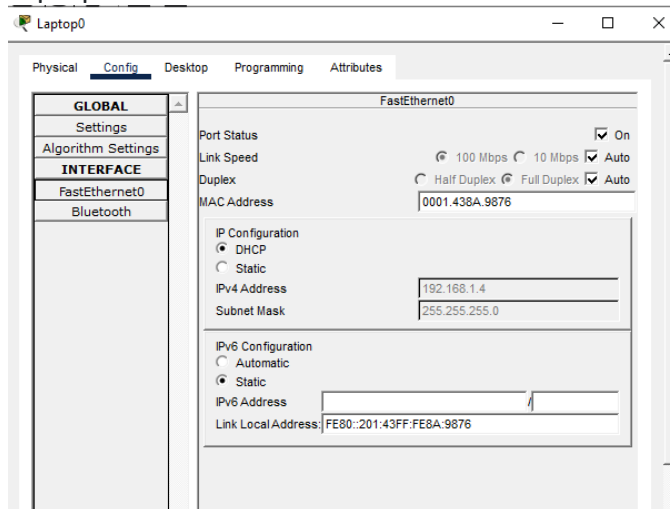


PC2

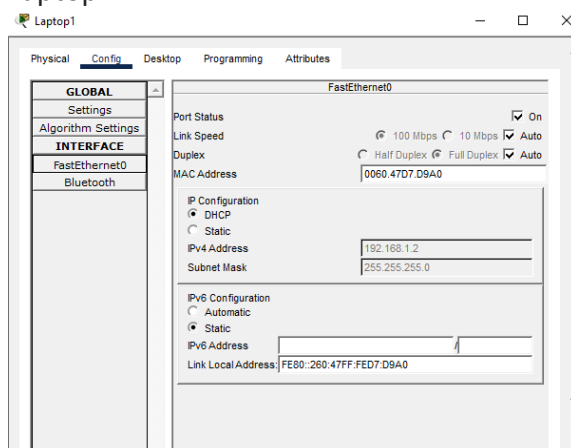


Not get IP, because it try to connect last. And Last IP was assign. With Maximum connection.

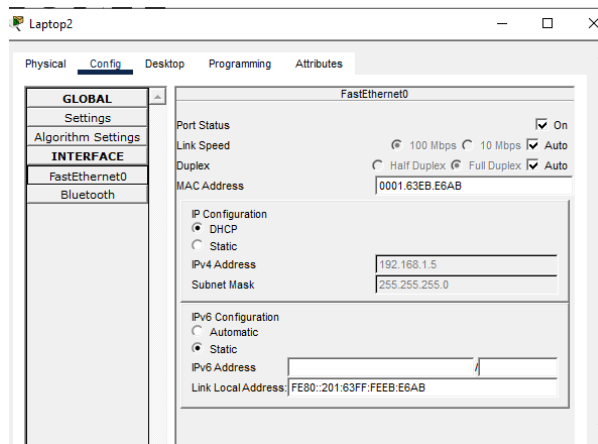
Laptop0



Laptop1



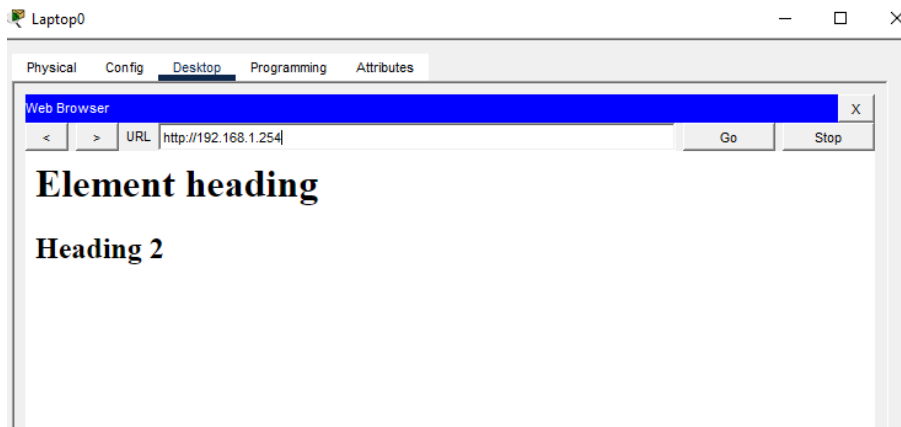
Laptop2



To Check DNS Resolver is working

Go to Laptop 2(Any device) → Desktop → Web Browser →

URL = 192.168.1.254



URL = Anudeep.com

